

Heat-Kernel Λ -Distributions on $U(1)$

Casimir filtering, endpoint behavior, and alias-free quadrature

Proof and computational verification note

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Outcome. Three diagonal $U(1)$ representations of dimensions 1, 2, 3 were checked at five heat times for all partitions through total degree five. Direct quadrature against the heat kernel agrees with the weight/Casimir formula in all **285** cases. The maximum absolute error is 5.73×10^{-14} .

1 Heat measure and normalization

Identify $U(1)$ with $e^{i\theta}$, $0 \leq \theta < 2\pi$, and choose the Laplacian normalization for which the character $e^{iq\theta}$ has Casimir q^2 . Brownian motion with generator $\frac{1}{2}\Delta$ has density relative to normalized Haar measure

$$k_t(\theta) = \sum_{q \in \mathbb{Z}} e^{-q^2 t/2} e^{-iq\theta}. \quad (1)$$

Orthogonality immediately yields

$$\int_{U(1)} e^{iw\theta} k_t(\theta) d\theta / (2\pi) = e^{-w^2 t/2}. \quad (2)$$

This fixes the otherwise convention-dependent factor in the Casimir exponent.

Let V be the diagonal representation with integer weights $a_1, \dots, a_d$,

$$\rho(e^{i\theta}) = \text{diag}(e^{ia_1\theta}, \dots, e^{ia_d\theta}). \quad (3)$$

For $W_\tau = \bigotimes_j \text{Sym}^{\tau_j} V$, write

$$W_\tau = \bigoplus_{w \in \mathbb{Z}} M_\tau(w) \mathbb{C}_w. \quad (4)$$

Combining (2)–(4) proves the coefficient formula

$$[\mathbf{t}^\tau] \mathcal{S}_t = \sum_{w \in \mathbb{Z}} M_\tau(w) e^{-w^2 t/2}. \quad (5)$$

This is the compact-group heat formula in its simplest exact form: each irreducible weight is attenuated by its Casimir eigenvalue.

2 Endpoints and interpretation

At $t = 0$, every exponential in (5) is one, hence

$$[\mathbf{t}^\tau] \mathcal{S}_0 = \dim W_\tau, \quad \mathcal{S}_0(\mathbf{t}) = \prod_j (1 - t_j)^{-d}. \quad (6)$$

This is evaluation at the identity matrix. As $t \rightarrow \infty$, only weight zero survives:

$$\lim_{t \rightarrow \infty} [\mathbf{t}^\tau] \mathcal{S}_t = M_\tau(0) = \dim W_\tau^{U(1)}. \quad (7)$$

Thus (5) gives a literal spectral interpolation between dimension and Haar invariants. For a purely positive defining weight there are no positive-degree Haar invariants. Mixed positive and negative weights, however, can cancel and leave nontrivial weight-zero terms.

3 Representations and verification method

The tested weight lists are

Name	weights	dimension	purpose
defining	(1)	1	isolates the scalar Casimir r^2
weight pair	(1, -1)	2	creates many weight-zero invariants
mixed	(0, 1, -2)	3	combines a fixed line with asymmetric weights

For each representation, $t = 0, 0.25, 0.8, 3, 20$ and all 19 partitions through total degree five are tested.

The direct route uses 512 equally spaced angles. At each angle it constructs the matrix (3), computes complete characters from the matrix power traces, and integrates their product against the Fourier heat kernel (1), truncated at $|q| \leq 32$. The formula route never evaluates an angle or matrix: it builds the integer multiplicities $M_\tau(w)$ and applies (5).

Why the quadrature is especially strong

This is not a generic Monte Carlo comparison. Through total degree five the largest character frequency is at most $5 \max_j |a_j| = 10$. Multiplication by the truncated kernel produces frequencies at most 42. Since 512 exceeds twice this bandwidth, the periodic trapezoidal rule integrates the entire trigonometric polynomial exactly in exact arithmetic. At the smallest positive time, the first omitted heat coefficient is $e^{-33^2/8} < 10^{-59}$. The observed errors are therefore floating-point roundoff, not sampling noise or appreciable heat-kernel truncation.

At $t = 0$ the heat density is a delta distribution, so the code evaluates the identity matrix directly as in (6). This avoids approximating a singular endpoint with a finite Fourier sum.

4 Results

Representation and time	coefficient	quadrature	Casimir formula	result
defining, $t = 0.8$	(3)	0.0273237224	0.0273237224	PASS
weight pair, $t = 3$	(2, 1)	0.8925233825	0.8925233825	PASS
mixed, $t = 20$	(2, 2)	5.0003631994	5.0003631994	PASS

The last value is already close to the integer Haar multiplicity five, while the small positive correction records the surviving nonzero weights at finite time.

5 Scope and reproducibility

The computation verifies the compact heat formula for $U(1)$ with several representation dimensions and both one-variable and mixed coefficients. It does not by itself verify the branching rules proposed for $U(n)$, $SU(n)$, $O(n)$, or $USp(2n)$. Those require separate highest-weight and Casimir normalizations. The disconnected-group warning is also essential: Brownian motion from the identity in $O(n)$ remains in $SO(n)$ unless component-changing jumps are added.

Run `compact_u1_heat/verification.py`, or open `compact_u1_heat/notebook.py` in marimo.